

Nested-quotient temporal histories and homological flat bands in strong-coupling $SU(N)$ Hamiltonian lattice gauge theory

Master derivation, with machine-checked provenance

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Abstract

We study the Kogut–Susskind Hamiltonian for pure $SU(N)$ gauge theory on a periodic cubic spatial lattice at strong coupling. After selecting the complete gauge-invariant charge-odd one-plaquette eigenspace, fixed-order contributions are organised as ordered magnetic histories on the Haar–Gram quotient of reachable Wilson networks. The complete momentum dependence through second order is the oriented edge–face Gram operator $t_N u^2 B(k) B(k)^\dagger$ with $B = \partial_2^\dagger$ and

$$t_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)} > 0, \quad N \geq 3.$$

The projected spectrum is $\{0, t_N u^2 q, t_N u^2 q\}$ up to a common scalar, with $q = 4 \sum_j \sin^2(k_j/2)$; the flat carrier is $\ker \partial_2$, of exact dimension $L^3 + 2$.

This edition closes the one step earlier versions asserted. The channel weights d_ρ/N^2 that fix t_N were previously introduced by an isotropy hypothesis; here they are *derived* from the order-two Weingarten values (Theorem 6), so the all-rank second-order result is self-contained in the strict sense. Three further statements are new: a Bloch sum rule joining the 3×3 spectrum to the $(L^3 + 2)$ -dimensional carrier (Theorem 4); the identification of the vacuum-mediated route that separates the charge-even and charge-odd hopping at every rank (Proposition 9); and an exact obstruction certificate showing that the retained Γ /axis corpus cannot identify the disputed fourth-order off-axis coefficient (Theorem 15).

$SU(3)$ third-order and fourth-order statements remain conditional on supplied exact-arithmetic ledgers whose raw payloads are not reproduced here, and are labelled as such throughout. Every displayed result carries the name of the machine check that establishes it. All statements concern a formal finite-order projected sector: this is not a continuum glueball or mass-gap theorem.

1 Introduction

Hamiltonian strong-coupling expansions organise lattice gauge dynamics around the electric vacuum and its flux excitations. In three spatial dimensions a fundamental plaquette excitation carries one of three face orientations, and at second order neighbouring faces communicate through a shared link, so the projected problem resembles a three-orbital tight-binding model. The signs of those matrix elements are fixed by the orientation of the plaquette boundary and by charge conjugation, and the resulting interference is the central object here.

Flat bands generated by incidence or Gram matrices have a long history: local topology, line-graph constructions, compact localised states, and the need for noncontractible states on periodic systems are all established, and closely related closed-surface structures occur in two-form spin liquids. We do not claim that incidence flatness, cellular compact localised states, or homological counting are new in isolation.

The content is dynamical. Starting from non-Abelian $SU(N)$ Hamiltonian lattice gauge theory, exact strong-coupling matrix elements generate an oriented edge–face Gram operator in the charge-odd one-plaquette sector. Representation theory fixes its coefficient and the cubical chain complex fixes its kernel and spectrum. The separation is exact,

$$\begin{aligned} \text{colour dynamics} &\longrightarrow t_N, \\ \text{cellular geometry} &\longrightarrow BB^\dagger. \end{aligned}$$

What is proved, and what is conditional

Four statements are self-contained here.

1. For every $N \geq 3$ the shared-link channel weights are d_ρ/N^2 , derived from the order-two Weingarten values (Theorem 6). This was previously an isotropy *hypothesis*, and everything in Section 4 rests on it.

2. For every $N \geq 3$ the complete momentum dependence at order u^2 is $t_N u^2 B B^\dagger$ (Theorem 8).
3. The Bloch spectrum of $B B^\dagger$ is $\{0, q, q\}$ (Theorem 1), and on the periodic L^3 complex the zero-mode count is $L^3 + 2$ (Theorem 2). The two are the same statement, by a sum rule over the momentum grid (Theorem 4).
4. The retained Γ /axis corpus cannot identify the disputed fourth-order off-axis coefficient (Theorem 15).

Two families of statements are *conditional* on supplied exact-arithmetic derivation ledgers. Their raw class contractions, symbolic history certificates and microscopic rank-sweep artifacts are not included here, and repeated assertions in synthesis documents are not treated as independent reproductions.

5. For $SU(3)$, supplied ledgers report that all candidate third-order lifter classes vanish and report the scalar and shared-link coefficients of Section 5. If that census is exhaustive and the linked ledger complete, the effective Hamiltonian stays in the incidence algebra through order u^3 .
6. Supplied ledgers report the all-rank cube formula $\alpha_N = 640/[N(N^2 - 1)^3]$ and its equality with the complete axial coefficient for $N = 3, \dots, 12$. Conditional on those calculations, the $SU(3)$ carrier acquires nonzero axial dispersion at order u^4 . The complete off-axis operator remains open — and by Theorem 15 it cannot be closed by any Γ or axial datum.

These statements concern the effective Hamiltonian projected to the degenerate one-plaquette manifold. Virtual multi-plaquette states are retained in the resolvents, but the external spectral problem is not the full gauge-theory Hilbert space. At $k = 0$ the three incidence branches touch and the finite-volume separation collapses as L^{-2} . Nothing below proves convergence of the strong-coupling series, persistence under unrestricted higher-order operators, identification with the lightest physical glueball, or a continuum mass gap.

2 Hamiltonian and projected sector

Let $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^3$, $L \geq 3$, be a cubic spatial lattice with periodic boundary conditions. In units of the electric coupling the Kogut–Susskind Hamiltonian is

$$H_\beta = \frac{1}{2} \sum_{\ell} C_2(\ell) + \beta_N \sum_p \left(1 - \frac{1}{N} \text{Re Tr } U_p\right). \quad (1)$$

Dropping the additive constant,

$$H = H_E - uV, \quad V = \sum_p (\chi_p + \bar{\chi}_p), \quad u = \frac{\beta_N}{2N}, \quad (2)$$

with $\chi_p = \text{Tr } U_p$. For $SU(3)$, $u = \beta_3/6$. The definition $u = \beta_N/(2N)$, not the $SU(3)$ -specific equality, is the all-rank convention; the archived line $Y = 2\beta/3 = 4u$ is a definition-label erratum and the printed coefficients are already in u .

check: the printed towers are canonical-u: 4*Delta(3u/2) reproduces them verbatim

All quoted excitation energies use the linked, vacuum-subtracted operator $H_{\text{exc}}(u) = H(u) - E_{\text{vac}}(u)I$. This changes only the scalar ledger; incidence hopping and carrier shape are unaffected.

The electric vacuum $|0\rangle$ carries the trivial representation on every link. For $N \geq 3$ the normalised one-plaquette states of definite charge conjugation are

$$|p, \pm\rangle = \frac{\chi_p \pm \bar{\chi}_p}{\sqrt{2}} |0\rangle, \quad (3)$$

of unit norm by character orthogonality. Each of the four boundary links carries the fundamental or antifundamental representation, so

$$E_{F,N} = 4\left(\frac{1}{2}C_F\right) = 2C_F = \frac{N^2 - 1}{N}, \quad C_F = \frac{N^2 - 1}{2N}. \quad (4)$$

check: each channel gap is C.F + C.R/2, and the weights sum to one

For $SU(2)$ complex conjugation is an inner gauge transformation and the charge-odd state vanishes; this is why the construction begins at $N = 3$. The same fact appears again in Section 4: $\Lambda^2 F$ is the singlet exactly at $N = 2$, which is where t_N vanishes.

We work inside the charge-odd fundamental Wilson-loop cyclic sector with trivial electric centre flux around each torus cycle. Let P_- be the full $E_{F,N}$ spectral projector there. For $L \geq 3$ its range is the charge-odd one-plaquette span, since a reduced contractible fundamental loop at that energy has four distinct links and is an elementary plaquette. This sector qualification also excludes the accidentally degenerate straight wrapping loop at $L = 4$. With $Q = 1_- - P_-$ the reduced resolvent crosses no omitted $E_{F,N}$ eigendirection, and degenerate perturbation theory gives

$$H_{\text{eff},-} = P_- H_E P_- + uH_1 + u^2 H_2 + u^3 H_3 + \dots \quad (5)$$

with every H_r linked, vacuum-subtracted, and carrying the appropriate Q -space resolvents.

3 Oriented incidence and the exact carrier

The periodic cubical complex has the cellular chain sequence $C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1$ with $\partial_2\partial_3 = 0$. One-plaquette amplitudes live in C_2 and their signed leakage into links is $\partial_2 c$: a cellular closed-surface condition, not the microscopic vertex Gauss law, which the Wilson-loop basis already satisfies through $\partial_1\partial_2 = 0$.

Writing $d_j = e^{ik_j} - 1$ and $q(k) = \sum_j |d_j|^2 = 4 \sum_j \sin^2(k_j/2)$, the face-to-link Bloch incidence matrix is

$$B(k) = \partial_2(k)^\dagger = \begin{pmatrix} d_2 & -d_1 & 0 \\ d_3 & 0 & -d_1 \\ 0 & d_3 & -d_2 \end{pmatrix}. \quad (6)$$

Theorem 1 (Incidence factorisation). *With $w(k) = (d_3, -d_2, d_1)^\top$,*

$$BB^\dagger = qI - ww^\dagger, \quad B^\dagger w = 0, \quad \|w\|^2 = q,$$

and therefore $\text{spec}(BB^\dagger) = \{0, q, q\}$. With $S = BB^\dagger - 4I$ the signed shared-link adjacency, $\text{spec } S(k) = \{-4, q - 4, q - 4\}$.

Proof. Multiplying (6) by its adjoint gives the first identity entry by entry. Since $\|w\|^2 = q$, the rank-one operator $qI - ww^\dagger$ annihilates w and acts as qI on its orthogonal complement. Subtracting $4I$ gives the adjacency spectrum. $\square$

check: `B Bdagger = q I - d conj(d)† for the curl incidence`

For $k \neq 0$ the carrier is one-dimensional and represented by $w(k)$. At Γ , $B = 0$ and all three branches meet; the normalised Bloch eigenvector has no direction-independent limit there, which is the familiar singular-flat-band mechanism behind the incompleteness of a purely compact localised basis.

Theorem 2 (Finite-torus carrier). *For the periodic cubic cellulation of T^3 , $\ker \partial_2 = \text{im } \partial_3 \oplus H_2$ with $\dim \text{im } \partial_3 = L^3 - 1$ and $\dim H_2 = 3$, so $\dim \ker \partial_2 = L^3 + 2$.*

Proof. There are L^3 three-cells and no four-cells, and $b_3(T^3) = 1$, so $\text{rank } \partial_3 = L^3 - 1$ by rank-nullity. Discrete Hodge decomposition gives $\ker \partial_2 = B_2 \oplus H_2$ with $\dim H_2 = b_2(T^3) = 3$. $\square$

The count is verified over $\mathbb{Z}$ rather than re-derived from the formula: the complex is built from the boundary formulas of Appendix D, rank over $\mathbb{F}_p$ bounds the kernel from below, and the exhibited cycles — every elementary cube boundary together with three wrapping sheets — bound it from above. The two bounds meet at $L = 1, \dots, 5$.

check: `dim Z.2 = L3 + 2 by rank, not by re-arranging the formula`

check: `cube boundaries and three wrapping sheets SPAN Z.2`

Remark 3. $L^3 + 2$ is the nullity of the down boundary Laplacian, not a Betti number; the Betti contribution is the final 3. The restriction $L \geq 3$ belongs to the twelve-neighbour Bloch adjacency, where $x+e$ and $x-e$ coincide at $L = 2$. The chain-level count has no such restriction and holds at $L = 1, 2$ as well.

check: `the L3+2 count is chain-level, not the Bloch convention`

The following sum rule joins Theorems 1 and 2, which are otherwise two unrelated statements: a 3×3 spectrum and an $(L^3 + 2)$ -dimensional space.

Theorem 4 (Bloch-chain bridge). *On the periodic L^3 complex,*

$$\sum_k (3 - \text{rank } B(k)) = L^3 + 2,$$

the sum running over the L^3 allowed momenta, with nullity profile 3 at Γ exactly once and 1 at each of the other $L^3 - 1$ momenta.

Proof. $B(0) = 0$ contributes 3. For $k \neq 0$ some $d_j \neq 0$, and Theorem 1 gives $\text{rank } B(k) = 2$, contributing 1 each. Summing gives $3 + (L^3 - 1) = L^3 + 2$. $\square$

check: `the Bloch and chain routes to the carrier agree`

Remark 5. The two decompositions produce the same two numbers from different objects. The 3 is the triple degeneracy of $B(0) = 0$, not $b_2(T^3)$; the $L^3 - 1$ counts non- Γ momenta, not cubes. Their agreement is a consistency check on the whole construction, not a restatement.

3.1 The wrapping sheets are cycles, not harmonic

Theorem 2 exhibits its three H_2 generators explicitly as wrapping sheets $s_{(ij)} = \sum_{x: x_m=0} [x; i, j]$. These satisfy $\partial_2 s = 0$ exactly, so they are cycles and they span the quotient $\ker \partial_2 / \text{im } \partial_3$. They are *not* harmonic: $\partial_3^\dagger s \neq 0$, and

$$\frac{\langle s, \partial_3 \partial_3^\dagger s \rangle}{\|s\|^2} = 2 \quad \text{exactly, for every } L \geq 2. \quad (7)$$

check: `FINDING: the wrapping sheets are cycles but NOT harmonic`

The distinction matters because H_2 is used in two senses: the quotient in Theorem 2, which the sheets span, and the harmonic subspace $\ker \partial_2 \cap \ker \partial_3^\dagger$, which they do not lie in. Nothing downstream breaks —

Proposition 14 rests on $B^\dagger w = 0$, which covers all of $\ker \partial_2$ — but a real-space statement phrased for the harmonic subspace does not cover the generators this construction exhibits.

4 Exact all-rank dynamics at second order

Two plaquettes sharing a link leave six nonshared fundamental links after the external contractions. On the shared link the fusion channels depend on the relative induced orientation,

$$F \otimes \bar{F} = \mathbf{1} \oplus \text{Adj}, \quad F \otimes F = \Lambda^2 F \oplus \text{Sym}^2 F, \quad (8)$$

with dimensions and quadratic Casimirs

$$\begin{aligned} d_1 &= 1, & C_1 &= 0, \\ d_{\text{Adj}} &= N^2 - 1, & C_{\text{Adj}} &= N, \\ d_{\Lambda^2 F} &= \frac{N(N-1)}{2}, & C_{\Lambda^2 F} &= \frac{(N+1)(N-2)}{N}, \\ d_{\text{Sym}^2 F} &= \frac{N(N+1)}{2}, & C_{\text{Sym}^2 F} &= \frac{(N-1)(N+2)}{N}. \end{aligned} \quad (9)$$

The six nonshared half-links contribute $3C_F$ and the fused link $C_\rho/2$, against an external one-plaquette energy $2C_F$, so the resolvent denominator is $C_F + C_\rho/2$ exactly.

check: each channel gap is $C_F + C_R/2$, and the weights sum to one

4.1 The channel weights are a theorem

Earlier editions assigned the channel projector squared norm d_ρ/N^2 by an isotropy hypothesis, and every statement in this section rests on that assignment. It is not a hypothesis.

Theorem 6 (Shared-link channel weights). *In the normalised two-plaquette state, the weight of the shared-link channel ρ is exactly d_ρ/N^2 , for both orientation families and every $N \geq 3$.*

Proof. Two steps. First the six nonshared links collapse. Each plaquette contributes a product of three independent Haar links, and a product of independent Haar matrices is Haar, so the amplitude is $\text{Tr}(AU) \text{Tr}(BU^{\pm 1})$ with A, B independent Haar and U the shared link. Integrating A and B contributes δ/N each and leaves a pure degree-(2, 2) moment of U .

Second, two such moments settle both families. With the order-two Weingarten values $\text{Wg}(e) = 1/(N^2 - 1)$ and $\text{Wg}((12)) = -1/(N(N^2 - 1))$ — the inverse of the S_2 Gram matrix $\begin{pmatrix} N^2 & N \\ N & N^2 \end{pmatrix}$ — the index sums give

$$\begin{aligned} M_{\text{direct}} &= \sum_{ijkl} \langle |U_{ij}|^2 |U_{kl}|^2 \rangle = N^2, \\ M_{\text{cross}} &= \sum_{ijkl} \langle U_{ij} U_{kl} \bar{U}_{il} \bar{U}_{kj} \rangle = N. \end{aligned} \quad (10)$$

For the like family, $\text{Tr}(AU) \text{Tr}(BU)$ decomposes into $\text{Sym}^2 \oplus \Lambda^2$ by symmetrising the two index pairs jointly, and the two squared norms are $(M_{\text{direct}} \pm M_{\text{cross}})/2$. Normalising,

$$n_{\text{Sym}^2} = \frac{N+1}{2N} = \frac{d_{\text{Sym}^2 F}}{N^2}, \quad n_{\Lambda^2} = \frac{N-1}{2N} = \frac{d_{\Lambda^2 F}}{N^2}.$$

For the mixed family the singlet component of $U_{ij} \bar{U}_{lk}$ is $\delta_{il} \delta_{jk}/N$, of squared norm 1 against $M_{\text{direct}} = N^2$, so $n_1 = 1/N^2 = d_1/N^2$ and $n_{\text{Adj}} = 1 - 1/N^2 = d_{\text{Adj}}/N^2$. $\square$

check: the shared-link weights are Weingarten, not an isotropy assumption

Remark 7. The derivation imports nothing: the Weingarten pair is the inverse of the S_2 Gram matrix and the index sums in (10) are explicit. It is also independent of the rest of this program, which is why it converts the all-rank second-order result from conditional to self-contained. Appendix B records the index sums.

With Theorem 6 the resolvent weight is

$$w_\rho = -\frac{d_\rho/N^2}{C_F + C_\rho/2}, \quad (11)$$

and summing within the two orientation families,

$$A_N = w_1 + w_{\text{Adj}} = -\frac{2N^3}{(N^2 - 1)(2N^2 - 1)}, \quad (12)$$

$$B_N = w_{\Lambda^2} + w_{\text{Sym}^2} = -\frac{4N(N^2 - 2)}{(N^2 - 1)(4N^2 - 9)}. \quad (13)$$

check: the four channel weights follow from dimension and Casimir

check: A_N and B_N are the channel sums, not transcriptions

4.2 The shared-link law

Theorem 8 (All-rank shared-link law). *For every integer $N \geq 3$,*

$$H_{\text{eff},-}^{(N)}(k, u) = E_{\text{flat},N}(u)I + t_N u^2 B(k)B(k)^\dagger + O(u^3),$$

with $E_{\text{flat},N}$ momentum independent and

$$t_N = B_N - A_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)} > 0.$$

The three bands through second order are $E_{\text{flat},N}(u)$ and $E_{\text{flat},N}(u) + t_N u^2 q(k)$ twice.

Proof. The diagonal and within-plaquette terms are scalars in face-orbital space. Every surviving second-order inter-plaquette process uses one shared link — see Proposition 9 for why the routes that do not are

absent from this sector — and its relative orientation is the incidence product appearing in $S = BB^\dagger - 4I$. The contributions (12) and (13) therefore assemble as $d_N I + t_N S$; absorbing $-4t_N$ into the scalar and applying Theorem 1 gives the bands. Positivity: for $N \geq 3$ every denominator factor is positive and $N^2 - 4 > 0$. $\square$

check: $t_N = B_N - A_N$

check: $t_N > 0$ for $N \geq 3$

Two consequences,

$$t_3 = \frac{5}{612}, \quad t_N = \frac{1}{4N^3} - \frac{1}{16N^5} - \frac{77}{64N^7} - \frac{1021}{256N^9} + \dots \quad (14)$$

check: $t_2 = 0$ and $t_3 = 5/612$

check: large- N expansion of t_N through $1/N^9$

4.3 The vacuum-mediated route

The proof of Theorem 8 asserts that only shared-link routes survive. That is false in general and true here, for a reason worth stating: it is a charge-conjugation selection rule, and omitting it is a recorded error, not a hypothetical one.

Proposition 9 (Vacuum-mediated route). *The route $\langle p'|V|0\rangle\langle 0|V|p\rangle/E_0$ connects any pair of plaquettes, sharing a link or not, and contributes $1/C_F$ per pair. It vanishes identically in the charge-odd sector, because V is charge-even and $|0\rangle$ is charge-even, so $\langle 0|V|p, -\rangle = 0$. Consequently the charge-odd hopping $t_N = B_N - A_N$ carries no such term, while the charge-even hopping does:*

$$\ell_N = A_N + B_N + \frac{1}{C_F} = -\frac{2N(3N^2 - 5)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)}.$$

check: $\text{ell}_N = A_N + B_N + 1/C_F$, the vacuum-mediated route at every rank

Remark 10. At $N = 3$, $A_3 + B_3 = -481/612$ and $1/C_F = 3/4$, giving $\ell_3 = -11/306$. The first of these is exactly the value once published for the charge-even hopping and later superseded; the correction is precisely the omitted route. Stating the selection rule turns a one-rank erratum into an all-rank formula.

4.4 Finite-volume width

Let $q_{\max}(L)$ be the largest sampled q on the periodic L^3 grid. Then

$$q_{\max}(L) = \begin{cases} 12, & L \text{ even,} \\ 12 \cos^2(\pi/2L), & L \text{ odd,} \end{cases} \quad (15)$$

because only at even L does some k_j reach π exactly. The width of the full retained charge-odd manifold is

$W_{N,L}^{(-)} = q_{\max}(L)t_N u^2 + O(u^3)$, so for even L and in the thermodynamic limit $[u^2]W^{(-)} = 12t_N \sim 3/N^3$. This concerns the stiffness of the dispersive doublet, not motion of the lowest branch, which is flat at this order.

check: the zone maximum of q is 12 only at even L

The smallest nonzero incidence eigenvalue is $q_{\min} = 4 \sin^2(\pi/L)$, attained by one unit of momentum in a single direction.

check: $q_{\min}$ on the L -torus grid is $4 \sin^2(\pi/L)$

4.5 The orientation signs are essential

Removing the orientation signs replaces each $e^{ik} - 1$ by $e^{ik} + 1$ and gives an unsigned incidence $\mathcal{N}(k)$ with $A + 4I = \mathcal{N}\mathcal{N}^\dagger$. Its adjacency spectra at Γ and R are $\{12, 0, 0\}$ and $\{-4, -4, -4\}$, which share no eigenvalue, so no level can be momentum independent. The flat branch is therefore not a generic consequence of three face orbitals or cubic symmetry.

This control is not hypothetical: the unsigned spectrum is the charge-even sector's, and the two spans are the two bandwidths. The signed span is $8 - (-4) = 12$, giving the charge-odd width $12t_- = 5/51$ at $N = 3$; the unsigned span is $12 - (-4) = 16$, giving the charge-even width $16/\ell_3 = 88/153$.

check: the two band spans ARE the two incidence spectra

5 $SU(3)$ through third order

Everything in this section is conditional on supplied exact-arithmetic ledgers. The within-plaquette charge-odd tower contributes $8/3 + u + u^2/2 + 7u^3/32$.

Reported computational result 11 (Third-order ledger). The supplied ledgers report, in exact rationals,

$$b_3 = \frac{1975}{124848}, \quad \text{leak}_3 = -\frac{12331}{249696},$$

and the assembled flat scalar $d_3 = \frac{7}{32} + 12 \text{leak}_3 - 4b_3 = -\frac{109151}{249696}$.

check: $d_3 = 7/32 + 12 \cdot \text{leak}_3 - 4 \cdot b_3$

check: leak_3 is assembled from the domino diagonal and the vacuum piece

Conditional corollary 12 (Third-order factorisation). If the lifter census is exhaustive and the linked ledger complete, then

$$H_{\text{eff},-}(k, u) = E_{\text{flat}}(u)I + \tau(u)B(k)B(k)^\dagger + O(u^4),$$

$$E_{\text{flat}}(u) = \frac{8}{3} + u + \frac{11}{306}u^2 - \frac{109151}{249696}u^3, \\ \tau(u) = \frac{5}{612}u^2 + \frac{1975}{124848}u^3.$$

check: E.flat and t(u) carry the ledger coefficients

The second-order assembly is $\frac{1}{2} + 12\text{leak}_2 - 4t_-$ with $\text{leak}_2 = -11/306$, giving the charge-odd diagonal $7/102$ and then the flat scalar $11/306$.

check: d.- = $1/2 + 12*\text{leak}_2$, and $\text{leak}_2 = -11/306$

5.1 One assembly, both sectors, both orders

The scalar ledger is not four separate facts. Writing λ for the adjacency eigenvalue and $\text{tower}_{r,s}$ for the within-plaquette term in canonical u ,

$$E_s(\lambda, r) = \text{tower}_{r,s} + 12\text{leak}_{r,s} + \lambda t_{r,s}, \quad (16)$$

with $\lambda \in \{-4, 8\}$ in the charge-odd sector (carrier and dispersive top) and $\lambda \in \{12, -4\}$ in the charge-even sector. This reproduces all eight registered band values across both sectors and both orders. The tower term is the certified coupling conversion $4\Delta(3u/2)$, so (16) takes no unregistered input.

check: one assembly formula gives every registered band value

Remark 13 (What the external comparison can and cannot do). Hamer's 1989 rest-frame series agrees with E_{flat} at every shared order. It cannot test τ . The hopping enters the spectrum only through $\tau(u)q(k)$ and $q(0) = 0$, so every Γ -point datum is blind to it; the one-parameter family $(b_3 + \varepsilon, \text{leak}_3 + \varepsilon/3)$ leaves d_3 and therefore the entire comparison unchanged. What the agreement pins is the combination $12\text{leak}_3 - 4b_3$. The $\lambda = 8$ band top separates them, via (16).

check: FINDING: no Gamma-point datum can constrain the hopping

6 What the homology protects

Proposition 14 (Boundary-factorised rigidity). *If an order- r Bloch correction has the form $H_r(k) = a_r I + b_r S(k) + B(k)M_r(k)B(k)^\dagger$, then for every $k \neq 0$, $H_r(k)w(k) = (a_r - 4b_r)w(k)$: the correction shifts the carrier without dispersing it.*

Proof. $B^\dagger w = 0$ and $Sw = -4w$. $\square$

In real space this is the orthogonality of the two cellular Laplacians, $L_2^\downarrow L_2^\uparrow = L_2^\uparrow L_2^\downarrow = 0$, so any polynomial in them acts on $\ker L_2^\downarrow \cap \ker L_2^\uparrow$ by its constant term. The proposition is not an unconditional all-orders statement: $\{BMB^\dagger\}$ is not a two-sided ideal in $\text{End}(C_2)$, and a general higher-order correction need not factor through links. Cubic symmetry alone permits carrier-projected fourth-order shapes built from

$$1, \quad q, \quad e_2, \quad \frac{4e_2}{q}, \quad \frac{e_3}{q}, \quad (17)$$

with $e_2 = \sum_{i < j} a_i a_j$, $e_3 = a_1 a_2 a_3$ and $a_i = 4 \sin^2(k_i/2)$.

7 The fourth-order boundary

Two supplied fourth-order records agree on the axial datum and on the appearance of mobility, and disagree on one off-axis mixed-gradient coefficient. This edition does not adjudicate them, and no fourth-order band or bandwidth is used above. What can be settled is whether the retained corpus *could* adjudicate them.

Theorem 15 (Obstruction certificate). *The two recorded off-axis coefficients differ, in the shape basis (17), by $4\Delta_C e_2$. On any one-dimensional axial cut two of the three a_i vanish, so e_2 is the zero polynomial there. Consequently no Γ -point or axial datum distinguishes the two records, at any precision. They differ at M and at R .*

check: FINDING: the retained Gamma/axis data cannot identify C.shp

Remark 16. The vanishing is polynomial, not numerical: this is non-identifiability, not a limitation of the data in hand. Re-anchoring cannot substitute, and neither record is preferred here. An off-axis contraction is the only decider.

On the axial cut itself the invariants collapse to $q = L(k) = 4 \sin^2(k/2)$ with $e_2 = e_3 = 0$, and dividing the raw cube-boundary numerator $(\alpha_N/4)L^2$ by $\|w\|^2 = q = L$ leaves a single power of L with coefficient $\alpha_3/4 = 5/48$.

check: on an axial cut the mixed invariants vanish and the norm divides

8 Scope

This work does not establish a four-dimensional infinite-volume Yang–Mills mass gap, a continuum glueball mass, convergence of the strong-coupling series to the continuum, regulator independence of the first dispersive order, the full $SU(3)$ fourth-order rest coefficient or off-axis shape, or an identification of any level here with a physical glueball. At Γ the three face orbitals transform as an axial vector of the cubic group, which with parity even and charge conjugation odd gives the retained-space label T_1^{+-} ; this is an operator-space assignment, not a spectral one.

The carrier is macroscopically degenerate at finite L , but its nearest incidence separation is $\Delta_L(u) = 4\tau(u) \sin^2(\pi/L) + O(u^4)$, which falls as L^{-2} and so provides no volume-uniform spectral isolation.

check: Delta.L = $4 \tau(u) \sin^2(\pi/L)$ is positive and falls as L^{-2}

9 Reproducibility

A portable standard-library Python program, `verify_core.py`, checks the rational coefficient ledger, the incidence identity, the chain condition $\partial_2\partial_3 = 0$, the finite-torus ranks, and the Weingarten derivation of Theorem 6. It has no dependencies and runs in about a fifth of a second. Two of its checks are stronger than stated elsewhere: the incidence identity is *decided* in exact Gaussian-rational arithmetic at rational torus points rather than checked to a residual, and the torus ranks come with an explicit kernel basis.

Every displayed result above names the machine check that establishes it. The checks live in the accompanying repository and each re-runs alone in about a second. Machine checks certify algebra relative to stated definitions and inputs; they do not replace a derivation, and they do not certify an uncompleted off-axis contraction — which is why Section 7 stops where it does.

A Channel weights in closed form

Substituting (9) into (11),

$$w_1 = -\frac{2}{N(N^2-1)}, \quad w_{\Lambda^2 F} = -\frac{N-1}{(N+1)(2N-3)},$$

$$w_{\Lambda \text{dj}} = -\frac{2(N^2-1)}{N(2N^2-1)}, \quad w_{\text{Sym}^2 F} = -\frac{N+1}{(N-1)(2N+3)}.$$

Their family sums are (12) and (13), and the difference has numerator $2N(N^2-4)$, which is the $SU(2)$ zero and the positivity for $N \geq 3$ at once.

B The Weingarten index sums

With $\text{Wg}(e) = 1/(N^2-1)$ and $\text{Wg}((12)) = -1/(N(N^2-1))$, and σ, τ running over S_2 ,

$$\langle U_{i_1 j_1} U_{i_2 j_2} \bar{U}_{i'_1 j'_1} \bar{U}_{i'_2 j'_2} \rangle$$

$$= \sum_{\sigma, \tau} \text{Wg}(\sigma\tau^{-1}) \prod_a \delta_{i_a i'_{\sigma(a)}} \delta_{j_a j'_{\tau(a)}}.$$

Summing the delta products over all N^4 index quadruples,

$$M_{\text{direct}} = \text{Wg}(e)(N^4 + N^2) + 2 \text{Wg}((12))N^3 = N^2,$$

$$M_{\text{cross}} = 2 \text{Wg}(e)N^3 + \text{Wg}((12))(N^4 + N^2) = N.$$

Both are confirmed by explicit summation at $N = 3, 4, 5$.

C The $SU(3)$ coefficient ledger

Order	Quantity	Value
u^0	electric plaquette	$8/3$
u^1	scalar	1
u^2	$BB^\dagger$ coefficient	$5/612$
u^2	flat scalar	$11/306$
u^3	$BB^\dagger$ coefficient	$1975/124848$
u^3	per-neighbour leakage	$-12331/249696$
u^3	flat scalar	$-109151/249696$

check: the manuscript's $SU(3)$ ledger is this registry, value by value

The u^1 row is $SU(3)$ -only rather than an $SU(3)$ specialisation: it is $-\langle p, -|V|p, - \rangle = 1$, which needs the ϵ channel and vanishes for $N \geq 4$. The other six rows specialise all-rank statements.

D Finite-volume chain calculation

For $i < j$,

$$\partial_2[x; i, j] = [x + e_i; j] - [x; j] - [x + e_j; i] + [x; i],$$

$$\begin{aligned} \partial_3[x; 1, 2, 3] &= [x + e_1; 2, 3] - [x; 2, 3] \\ &\quad - [x + e_2; 1, 3] + [x; 1, 3] \\ &\quad + [x + e_3; 1, 2] - [x; 1, 2]. \end{aligned}$$

Substituting the second into the first cancels every edge pairwise, so $\partial_2\partial_3 = 0$ over $\mathbb{Z}$.

check: `d.2 d.3 = 0` on the built complex

On the torus $\ker \partial_3$ is generated by the sum of all oriented cubes, so $\text{rank } \partial_3 = L^3 - 1$.

check: `rank d.3 = L3 - 1` on the built complex